

1 Notation & Perturbation Method

Big $\mathcal{O}$: $f(z) = \mathcal{O}(g(z))$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} \text{ bdd} \Leftrightarrow |f(z)| \leq A|g(z)|$ as $z \rightarrow z_0$

Small o : $f(z) = o(g(z))$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = 0 \Leftrightarrow f(z) \ll g(z)$ as $z \rightarrow z_0$

Asymptotic $\sim$: $f(z) \sim g(z)$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = 1$ **ps:** $e^{-x} = o(x^{-n}), \forall n > 0$ as $x \rightarrow \infty$

Asymptotic Sequence: Seq $\{\varphi_n(z)\}$ is asy. if $\forall n \in \mathbb{N} \varphi_{n+1}(z) = o(\varphi_n(z))$ as $z \rightarrow z_0$

Asymptotic Expansion: $f(z) \sim \sum_{n=0}^M a_n \varphi_n(z)$ as $z \rightarrow z_0$ to M if $\phi_n(z)$ is asy. seq. and

$f(z) - \sum_{n=0}^N a_n \varphi_n(z) = o(\varphi_N(z)) = \mathcal{O}(\varphi_{N+1}(z)), \forall N \leq M$ as $z \rightarrow z_0$

$f(x, \varepsilon) = \sum_{n=0}^{\infty} a_n(x) \varepsilon^n$ uniformly in $x \in I$ as $\varepsilon \rightarrow 0$ if ε 与 x 无关 (i.e. sup $|f(x, \varepsilon) - \sum_{n=0}^N a_n(x) \varepsilon^n| = o(\varepsilon^N)$)

Singular|Regular - Perturbation Method: a. Let $\varepsilon = 0$ solve $f(x, 0) = 0$.

b.i. 如果 f 结构变了 (阶数降低), Let $x = \varepsilon^{-\alpha} X$, PDB 找 α . c. 假设 $X = a_0 + a_1 \varepsilon^b + a_2 \varepsilon^{2b} + \dots$ 代入求解

b.ii. 如果 f 结构不变/变, 但有重根 x_0 , Let $x = x_0 + \varepsilon^\alpha X$, PDB 找 α . $\uparrow \beta$ 简化后的 X 方程得出

Principle of dominant balance (PDB): For Singular Perturbation $p(x, \varepsilon) = 0$, Let $x = \varepsilon^\alpha X$, 代入选择 α :

对于 ε 来说 a. 至少有两项阶数相同且最低; b. 其他项阶数均更高. (即可忽略)

2 Integral By Parts, Laplace's Method & Method of Steepest Descents

Taylor: $f(x) = \sum_{n=0}^N \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n + \frac{f^{(N)}(\xi)}{N!} (x - x_0)^N \Rightarrow f(x) \sim \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$ as $x \rightarrow x_0$

Exp. Integral: $E_1(x) = \int_x^\infty \frac{e^{-t}}{t} dt$, $\{x \rightarrow 0 \text{ 发散}\}$ $E_1(x) \sim e^{-x} \sum_{n=0}^{\infty} \frac{(-1)^n}{x^{n+1}}$, $x \rightarrow \infty$

Gamma Func: $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$ $\Gamma(x+1) = x\Gamma(x)$ and $\Gamma(n) = (n-1)!$ $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Incomplete Gamma Func: **Lower:** $\gamma(a, x) = \int_0^x t^{a-1} e^{-t} dt$ **Upper:** $\Gamma(a, x) = \int_x^\infty t^{a-1} e^{-t} dt$

$\gamma(a, x) = \Gamma(a) - \Gamma(a, x)$ $\Gamma(a, x) \sim e^{-x} x^{a-1} \sum_{n=0}^{\infty} \frac{\Gamma(a)}{\Gamma(a-n)} x^{-n}$, $x \rightarrow \infty$

Laplace's Method: For $I(x) = \int_a^b e^{xf(t)} g(t) dt$ as $x \rightarrow \infty$ $\frac{f''(t_0)}{2} (t - t_0)^2 + \dots$

1. If $f(t)$ has unique max at $t = t_0 \in (a, b) \Rightarrow I \sim \int_{-\infty}^{\infty} e^{xf(t_0) + \frac{f''(t_0)}{2} (t - t_0)^2 + \dots} g(t_0) dt$

(留 n 项非常数展开为 n term approximation) (ps: $\max \leftrightarrow f'(t_0) = 0, f''(t_0) < 0$)

Useful Tool: Now: $I \sim e^{xf(t_0)} \int_{-\infty}^{\infty} e^{x[\frac{f''(t_0)}{2} (t - t_0)^2 + \dots]} g(t_0) dt \Rightarrow \text{Let } \tau^2 = -[\frac{1}{2} f''(t_0)] (t - t_0)^2 + \dots$

Assume $t = \tau + b\tau^2 + c\tau^3 + \dots$ as $\tau \rightarrow 0 \Rightarrow$ 代入换元式子求系数 $\Rightarrow dt = (1 + 2b\tau + 3c\tau^2 + \dots) d\tau$

2. **Exact Substitution:** 由 1 的基础, 得到 $f(t)$ 在 t_0 处的展开式, 换元令 $f(t)$ 就等于展开式前几项求解.

3. If $g = g(t, x): e^{xf(t)} g(t, x) = e^{xf(t) + \ln(g(t, x))}$, 看 $xf(t) + \ln(g(t, x))$ 的最大值, 类似 1 处理.

- If $t_0 = f(x)$, 换元令 $t = f(x)$.

Method of Steepest Descents: For $I(x) = \int_C e^{xf(z)} g(z) dz$ as $x \rightarrow \infty$. f 容易避开 singular point

If z_0 is saddle point: $f'(z_0) = 0, f''(z_0) \neq 0$ $f(z)$ Taylor at z_0 : $f(z) = f(z_0) + \frac{f''(z_0)}{2} (z - z_0)^2 + \dots$

1. Let $z = z_0 + e^{i\theta} y$ (绕 $f''(z_0) = e^{2i\theta}$ 为负实数) $\rightarrow I(x) \sim \int_{-\infty}^{\infty} e^{x[f(z_0) - \frac{|f''(z_0)|}{2} y^2 + \dots]} g(z_0) e^{i\theta} dy$

2. If f 不容易避开 singular point: a. $\text{Im}(f(z)) = \text{Im}(f(z_0))$ and $z_0 \in C_0$ to find steepest descent path C_0

3. b. 画图看哪些 singular points 在 $C \rightarrow C_0$ 变形路径上, 计算 residues. c. $I = 1$ 部分 + $2\pi i \sum$ residues.

4. **More than one saddle point:** i. 如果通过积分区域可以舍弃, 则舍弃 ii. saddle points: z_k , 取能让 $\text{Re}(f(z_k))$ 最小的.

iii. 如果多个 saddle points 都满足 $\text{Re}(f(z_k))$ 最小, 则都要计算后叠加.

5. **Exact Substitution:** 由 1 的基础, 得到 $f(z)$ 在 z_0 处的展开式, 换元令 $f(z)$ 就等于展开式前几项求解.

6. **Remark:** 类似 Laplace 3 一样, 可以对 $xf(x)$ 一个整体处理, 必要时 $z_0 = f(x) \rightarrow$ 换元 $z = f(x)$.

Residue: If z_0 pole of order m of f , $\Rightarrow \text{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]$

Method of Stationary Phase: $\cos(x f(\theta)) \rightarrow \text{Re}(e^{ix f(\theta)})$ 回到 Steepest Descents, 同理 $\sin(x f(\theta))$

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3 Perturbation Methods in ODEs

Regular Perturbation: For $L(y, \varepsilon) = 0$, let $y(x, \varepsilon) = y_0(x) + \varepsilon y_1(x) + \varepsilon^2 y_2(x) + \dots$

$\Rightarrow$ a. 代入 $\mathcal{O}(\varepsilon^n) \rightarrow$ b. Boundary/Initial Condition 求解 $y_n(x) \rightarrow$ a.

Singular Perturbation: e.g. $\varepsilon y'' + p(x)y' + q(x)y = 0, y(a) = A, y(b) = B, a < b$, 大致步骤:

1. Assume boundary layer at $x = a|b$. Let $x = a + \delta(\varepsilon)X$ or $x = b - \delta(\varepsilon)X$

2. Assume inner $Y(X) = y(x)$, chain rule 求代入内层 ODE $\rightarrow$ Domain balance 找出 $\delta(\varepsilon) = \varepsilon^\alpha$

Recall PDB: a. 至少有两项阶数相同且最低; b. 其他项阶数均更高.

3. Assume $Y = Y_0(X) + \varepsilon Y_1(X) + \varepsilon^2 Y_2(X) + \dots$ 代入求解 $\rightarrow$ 边界/初始条件求解 $Y_0(X)$

4. **Matching:** Consider $X \rightarrow \infty|x \rightarrow a|b$, If limits not exist (除非 X) $\rightarrow$ boundary layer 在另一个端点 $\rightarrow 1$

If limits exist, boundary layer 正确 $\rightarrow$ 渐进 outer (X) , sup inner 极限 (X) : 算不了保留 $X \rightarrow$ 和 X 的关系代入 $\rightarrow$ 比较系数.

Remark 1: Outer solution 只能用在 away from boundary layer 的区域 (包括边界点).

Remark 2: Inner solution 只能用在 boundary layer 附近的区域 (包括边界点).

Useful Lemma: For $\varepsilon y'' + p(x)y' + q(x)y = 0, y(a) = A, y(b) = B, a < b, p(x) \neq 0$ in $[a, b]$ then

Has boundary layer at $x = a$ if $p > 0$; Has boundary layer at $x = b$ if $p < 0$.

4 WKB Theory & Solving PDE

WKB (I): For $u'' + k^2 q(x)u = 0, k \gg 1$ we have: $u \sim A(x)e^{ik\phi(x)}$. Substitution can get: (q 里不能带 k)

If $q(x) < 0: u(x) \sim A_+ |q|^{-\frac{1}{4}} e^{-\frac{k}{2} \int_{x_0}^x \sqrt{|q(t)|} dt} + A_- |q|^{-\frac{1}{4}} e^{-\frac{k}{2} \int_{x_0}^x \sqrt{|q(t)|} dt}$ (Exp. Decay/Growth)

If $q(x) > 0: u(x) \sim A_+ |q|^{-\frac{1}{4}} e^{\frac{ik}{2} \int_{x_0}^x \sqrt{q(t)} dt} + A_- |q|^{-\frac{1}{4}} e^{-\frac{ik}{2} \int_{x_0}^x \sqrt{q(t)} dt}$ (Oscillation)

Turning Point: If $q(x_0) = 0$, then x_0 is a WKB turning point. ps: 如果 $q(x)$ 含 k , 对其展开保留到 k^0 项

WKB (II): For $u'' + k^2 q(x)u = 0, k \gg 1$ Then:

If $q(x) < 0$: Assume $u \sim A(x)e^{ik\phi(x)}$ then: $u' = (A' + k\phi' A)e^{ik\phi}$ and

$u'' = (A'' + 2k\phi' A' + k\phi'' A + k^2 \phi'^2 A)e^{ik\phi}$ 代入并比较 $\mathcal{O}(k^2), \mathcal{O}(k)$ (No $\mathcal{O}(1)$)

If $q(x) > 0$: Assume $u \sim A(x)e^{ik\phi(x)}$ then: $u' = (A' + ik\phi' A)e^{ik\phi}$ and

$u'' = (A'' + 2ik\phi' A' + k\phi'' A - k^2 \phi'^2 A)e^{ik\phi}$ 代入并比较 $\mathcal{O}(k^2), \mathcal{O}(k)$ (No $\mathcal{O}(1)$)

Method of Characteristics|IC: For $u_t + c(x, t)u_x = r(x, t, u)$ with IC $u(x, t_0) = f(x)$:

1. In characteristic form: $\frac{dx}{dt} = r(x, t, u)$ on $\frac{dt}{dx} = c(x, t) \rightarrow$ 解 ODEs $\rightarrow u, t$ 各产生一个未知常数 A, B .

2. Assume characteristic curve (solution of $\frac{dx}{dt} = c(x, t)$) $x(t = t_0) = \xi$ SO $u(t = t_0) = f(\xi)$:

$\rightarrow$ get $u(t = t_0) = f(\xi)$ on $x(t = t_0) = \xi \rightarrow$ solve A, B .

3. Substitute A, B back to get u .

Method of Characteristics|BC: For $u_t + c(x, t)u_x = r(x, t, u)$ with BC $u(x_0, t) = f(t)$:

1. In characteristic form: $\frac{dx}{dt} = r(x, t, u)$ on $\frac{dt}{dx} = c(x, t) \rightarrow$ 解 ODEs $\rightarrow u, t$ 各产生一个未知常数 A, B .

2. Assume characteristic curve (solution of $\frac{dx}{dt} = c(x, t)$) $t(x = x_0) = \xi$ SO $u(x = x_0) = f(\xi)$:

$\rightarrow$ get $u(x = x_0) = f(\xi)$ on $t(x = x_0) = \xi \rightarrow$ solve A, B .

3. Substitute A, B back to get u .

Method of Characteristics|BC: For $u_t + c(x, t)u_x = r(x, t, u)$ with BC $u(x_0, t) = f(t)$:

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2. Assume characteristic curve (solution of $\frac{dx}{dt} = c(x, t)$) $t(x = x_0) = \xi$ SO $u(x = x_0) = f(\xi)$:

$\rightarrow$ get $u(x = x_0) = f(\xi)$ on $t(x = x_0) = \xi \rightarrow$ solve A, B .

3. Substitute A, B back to get u .

Special Cases of Method of Characteristics:

1. 如果 Characteristic curve 和 初始曲线/边界条件相交 $\rightarrow$ ill-posed & if not unique $\rightarrow$ no solution.

2. **For $u_t + c(u)u_x = 0$:** If an initial jump $u(x, t_0) = u_L$ for $x < b$ and $u(x, t_0) = u_R$ for $x > b$

If $c(u_L) < c(u_R) \rightarrow$ expansion wave; If $c(u_L) > c(u_R) \rightarrow$ shock wave.

3. **For $u_t + c(u)u_x = 0, u(x, t_0) = f(x)$:** The breaking time $t_b = t_0 + \min_{\xi} \frac{-1}{F'(\xi)}$ where $F(\xi) = c(f(\xi))$

Break time is $\frac{du}{dx} = \infty$ (i.e. 分母为 0) 时的最早时间. (通过 chain rule 得到) || If $\min_{\xi} F'(\xi) < 0 \rightarrow$ shock

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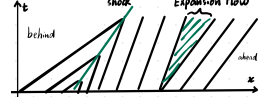
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Expansion: For $u_t + c(u)u_x = 0, u(x, t_0) = u_L|u_R$

1. 正常解 $\rightarrow \xi$ 边界值 $\xi_0 \rightarrow$ 特征曲线 $x(t; \xi)$ 在 ξ_0 边界情况:

$x = c(u_L)(t - t_0) + \xi_0$ 和 $x = c(u_R)(t - t_0) + \xi_0$

2. $\Rightarrow \frac{x - \xi_0}{t - t_0} = c(u)$ for $c(u_L)(t - t_0) + \xi_0 < x < c(u_R)(t - t_0) + \xi_0$
 $\Rightarrow u = c^{-1}[\frac{x - \xi_0}{t - t_0}]$



Conservation Law: $u_t + c(u)u_x = 0: u_t + [f(c(u))u]_x = 0$

Shock Wave u_L, u_R : For $u_t + c(u)u_x = 0, u(x, t_0) = u_L$ for $x < b$ and $u(x, t_0) = u_R$ for $x > b$.

1. 正常解 $\rightarrow$ In conservation form: $u_t + [f(c(u))u]_x = 0$

shock velocity $\frac{ds}{dt} = \dot{s} = \frac{[q]}{[\rho]}$ where $q = f(c(u))u$ and $\rho = u; [u] := u_{\text{ahead}} - u_{\text{behind}}$:

其中: $u_{\text{ahead}} = u_R, u_{\text{behind}} = u_L$

2. Solve $\dot{s}$. As shock starts at $x = b$ at $t = t_0, s(t = t_0) = b$ get unknown constant.

3. Then solution $u = u_L$ for $x < s(t)$ and $u = u_R$ for $x > s(t)$.

Shock Wave $f(x)$: For $u_t + c(u)u_x = 0, u(x, t_0) = f(x)$ with $t_b = t_0 + \min_{\xi} \frac{-1}{F'(\xi)}$ where $F(\xi) = c(f(\xi))$.

1. 正常解 $\rightarrow$ In conservation form: $u_t + [f(c(u))u]_x = 0$

shock velocity $\frac{ds}{dt} = \dot{s} = \frac{[q]}{[\rho]}$ where $q = f(c(u))u$ and $\rho = u; [u] := u_{\text{ahead}} - u_{\text{behind}}$:

其中: $u_{\text{ahead}} = \lim_{x \rightarrow s(t)^+} u(x, t), u_{\text{behind}} = \lim_{x \rightarrow s(t)^-} u(x, t)$ (which is calculated by characteristics method)

2. Solve $\dot{s}$. As shock starts at $x(t = t_b)$ at $t = t_b, s(t = t_b) = x(t = t_b)$ get unknown constant.

ps: i.e. $x(t = t_b) = \xi_*$ and $F(\xi_*) (t_b - t_0)$, 其中 ξ_* 是使得 $t_b = t_0 + \min_{\xi} \frac{-1}{F'(\xi)}$ 的 ξ .

Remark: If $u_t + c(u)u_x = 0$, 大体思路都不变, 除了 In conservation form: $u_t + [f(c(u))u]_x = r(x, t, u)$.

Remark: 可能 Expansion and Shock 冲突 $\rightarrow$ 原先 Shock 消失并留 Expansion 合并 $\rightarrow$ 寻找第一次 $t_b > 0$ 可能发生的时间

$\rightarrow$ 解新 s ; u_R, u_L is expansion f 值, shock 前的值 $\rightarrow$ 用原先 $s(t = t_b)$ 解新 s 未知数 $\rightarrow$ 新 u , 留 expansion f .

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7 Green’s Function

ps: To solve *forced equation*. 即 PDE RHS ≠ 0

Green's Function: For PDE $L u(x, t) = F(x, t)$ where L is *linear* differential operation with BCs and ICs.

1. Consider $G(x, t; \xi, \tau)$ satisfying $L G(x, t; \xi, \tau) = \delta(x - \xi) \delta(t - \tau)$ with BCs and ICs.
2. Solve $G \rightarrow G(x, t; \xi, \tau)$.
3. Then $u(x, t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} G(x, t; \xi, \tau) F(\xi, \tau) d\xi d\tau$.

ps: 可以拓展到更多的空间维度, e.g. $G(x, y, t; \xi, \eta, \tau) \rightarrow L G(x, y, t; \xi, \eta, \tau) = \delta(x - \xi) \delta(y - \eta) \delta(t - \tau)$

- How to solve: $L G(x, t; \xi, \tau) = \delta(x - \xi) \delta(t - \tau)$ with BCs and ICs. ↓

a. Fourier Transform (infinite space domain (x, t) , [2 dimensions])

1. Take Fourier transform in $x \rightarrow$ Get ODE $\rightarrow$ Solve ODE with $t \neq \tau \rightarrow$ Get $\overline{G}$ split by $t < \tau$ and $t > \tau$.
2. ICs $\rightarrow$ Get some unknown constants in $\overline{G}$.
3. Apply **Jump Condition** to ODE, $\int_{\tau-}^{\tau+} \cdot dt \rightarrow$ Get unknown constant.
4. Inverse Fourier $\rightarrow G(x, t; \xi, \tau)$.

b. Separation of Variables (finite space domain (x))

1. Need homogeneous BCs $\rightarrow$ Assume $t \neq \tau$ and $x \neq \xi \rightarrow$ Separation of variables $\rightarrow$ Get λ_n and $X_n(x)$.
2. Assume $G = \sum_n g_n(t) X_n(x) \rightarrow$ Substitute into PDE $L G = \delta(x - \xi) \delta(t - \tau)$.
3. Use orthogonality of $X_n \rightarrow$ Get $\delta(x - \xi) = \sum_n d_n X_n(x)$ and find d_n .
4. 通过前面 substitute 得到的对比 terms $\rightarrow$ Get ODE for $g_n(t) \rightarrow$ Solve ODE split by $t < \tau$ and $t > \tau$.
5. ICs $\rightarrow$ Get some unknown constants in $g_n(t) \rightarrow$ **Continuity** at $t = \tau \rightarrow$ some unknown constants.
6. Apply **Jump Condition** to ODE, $\int_{\tau-}^{\tau+} \cdot dt \rightarrow$ Get unknown constant $\rightarrow$ Get $g_n(t)$ and we solve it :)

Remark: 可以拓展为更多空间维度 $\rightarrow$ Separation of variables 后留空间不算到最后假设来找.
e.g. $G_t - \Delta G = \delta(x - \xi) \delta(y - \eta) \delta(t - \tau) \rightarrow \dots \rightarrow G = \sum_{m,n} g_{m,n}(t) X_n(x) Y_m(y) \rightarrow \dots$

c. High Variables / Laplace Transform

Remark: 如 $F(x, t)$ 给具体函数 $\rightarrow$ 考虑齐次 PDE $\rightarrow$ Eigenfunction $X_n(x) \rightarrow u = \sum_n g_n(t) X_n(x) \rightarrow$ RHS 含 x 的写为 $\sum_n d_n X_n(x) \rightarrow$ 比较系数 $\rightarrow$ 得到 $g_n(t)$ 的 ODE $\rightarrow$ Solve ODE $\rightarrow$ 最后得到 $u(x, t)$ i.e. 跳过求 G 的步骤, 直接用 X_n 展开 RHS

9 Appendix II

Useful Tools: $\int_0^L \sin^2(ax) dx = \frac{L}{2} - \frac{\sin(2aL)}{4a}$ $\int_0^L \cos^2(ax) dx = \frac{L}{2} + \frac{\sin(2aL)}{4a}$

$\int_0^L \sin(ax) \cos(ax) dx = \frac{\sin^2(aL)}{2a}$

8 Laplace Transform Table

$f(t)$	$\mathcal{L}\{f(t)\}$	$f(t)$	$\mathcal{L}\{f(t)\}$	$f(t)$	$\mathcal{L}\{f(t)\}$
1	$\frac{1}{s}$	t^n	$\frac{n!}{s^{n+1}}$	$f'(t)$	$s\overline{f}(s) - f(0)$
e^{at}	$\frac{1}{s-a}$	$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$	$\int_0^t f(\tau) d\tau$	$\frac{1}{s}\overline{f}(s)$
$\sin(at)$	$\frac{a}{s^2+a^2}$	$t^n \sin(at)$	$\frac{n!a}{(s^2+a^2)^{n+1}}$	$e^{at} f(t)$	s-shift $\overline{f}(s-a)$
$\cos(at)$	$\frac{s}{s^2+a^2}$	$t^n \cos(at)$	$\frac{n!s}{(s^2+a^2)^{n+1}}$	$f(t-a)u_a(t)$	t-shift $e^{-as}\overline{f}(s)$
$\sinh(at)$	$\frac{a}{s^2-a^2}$	$t^n \sinh(at)$	$\frac{n!a}{(s^2-a^2)^{n+1}}$	$\int_0^t f(t-\tau)g(\tau) d\tau$	Convolution $\overline{f}(s) \cdot \overline{g}(s)$
$\cosh(at)$	$\frac{s}{s^2-a^2}$	$t^n \cosh(at)$	$\frac{n!s}{(s^2-a^2)^{n+1}}$	$f^n(t): s^n \overline{f}(s) - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$	
$u_b(t)$	$\frac{e^{-bs}}{s}$	$\delta(t-b)$	e^{-bs}	$f * g = \int_0^t f(t-\tau)g(\tau) d\tau$	

ps: $u_b(t) = H(t-b) = \begin{cases} 0, & t < b \\ 1, & t \geq b \end{cases}$ $\int_{-\infty}^{\infty} \delta(t-b)\phi(t)dt = \int_b^-^{b+} \delta(t-b)\phi(t)dt = \phi(b)$

Hyperbolic Functions: $\sinh(x) = \frac{e^x - e^{-x}}{2}$ $\cosh(x) = \frac{e^x + e^{-x}}{2}$

$1 + \tanh^2(x) = \operatorname{sech}^2(x)$ $\coth^2(x) - 1 = \operatorname{csch}^2(x)$

ps: $1 + \tan^2(x) = \sec^2(x)$ $1 + \cot^2(x) = \operatorname{csc}^2(x)$